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Registration number: FA23-BSE-142

Q1

algorithm

Student age validator

```
package model;

/**
 *
 * @author user
 */
public class StudentAgeValidator {

    public boolean validateAge(int age) {

        if (age < 0) {
            throw new IllegalArgumentException("Age cannot be negative");
        }

        return age >= 5;
    }

}
```

Student age validator test

Positive test

```
package test;

/**
 *
 * @author user
 */

import model.StudentAgeValidator;

import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

public class StudentAgeValidatorTest {

    // Positive Test
    @Test
    void shouldAcceptValidAge() {

        StudentAgeValidator validator =
            new StudentAgeValidator();

        assertTrue(
            validator.validateAge(18)
        );

    }

    // Negative Test
    @Test
    void shouldRejectInvalidAge() {
```

Negative test

```

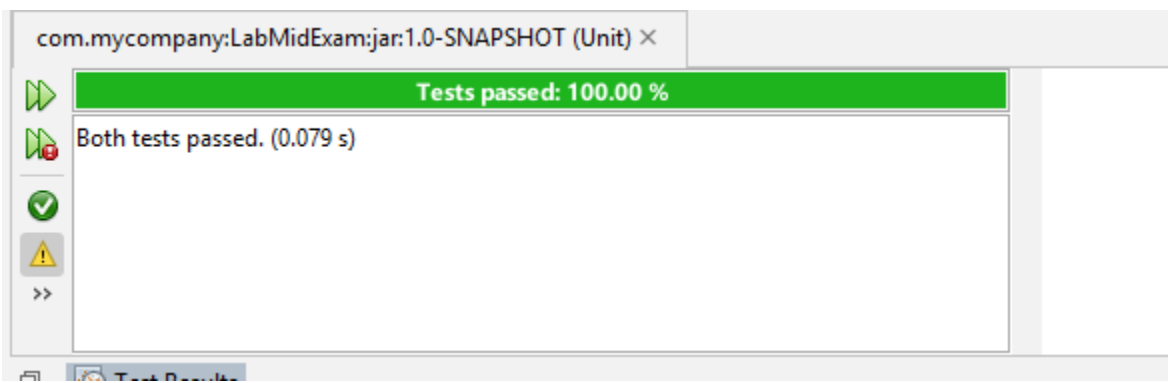
// Negative Test
@Test
void shouldThrowExceptionForNegativeAge() {

    StudentAgeValidator validator =
        new StudentAgeValidator();

    assertThrows(
        IllegalArgumentException.class,
        () -> validator.validateAge(-2)
    );
}
}

```

Output



Question 2

Case study hospital management system

Use case schedule appointment

Patient

```
package model;

/**
 *
 * @author user
 */

public class Patient {

    private String name;

    public Patient(String name) {

        this.name = name;

    }

    public String getName() {

        return name;

    }

}
```

Doctor

```
package model;

/**
 *
 * @author user
 */

public class Doctor {

    private String name;

    public Doctor(String name) {

        this.name = name;

    }

    public String getName() {

        return name;

    }

}
```

Unpacking ir

Appointment

```
package model;

/**
 *
 * @author user
 */

public class Appointment {

    private Patient patient;
    private Doctor doctor;

    public Appointment(
        Patient patient,
        Doctor doctor) {

        this.patient = patient;
        this.doctor = doctor;

    }

}
```

Appointment manager

```
package model;

/**
 *
 * @author user
 */
import java.util.ArrayList;
public class AppointmentManager {
    private ArrayList<Appointment>
        appointments =
            new ArrayList<>();
    public void scheduleAppointment(
        Patient patient,
        Doctor doctor) {

        appointments.add(
            new Appointment(
                patient,
                doctor
            )
        );
    }

    public ArrayList<Appointment>
    getAppointments() {
        return appointments;
    }
}
```

Appointment manager test

```

    */
    import model.*;
    import org.junit.jupiter.api.Test;
    import static org.junit.jupiter.api.Assertions.*;
    public class AppointmentManagerTest {
        @Test
        void shouldScheduleAppointment() {
            AppointmentManager manager =
                new AppointmentManager();
            Patient patient =
                new Patient(
                    "Ali"
                );
            Doctor doctor =
                new Doctor(
                    "Dr Ahmed"
                );
            manager.scheduleAppointment(
                patient,
                doctor
            );
            assertEquals(
                1,
                manager.getAppointments().size()
            );
        }
    }
}

```

Output

com.mycompany:LabMidExam:jar:1.0-SNAPSHOT (Unit) × com.mycompany:LabMidExam:jar:1.0-SNAPSHOT

Tests passed: 100.00 %

The test passed. (0.089 s)

Test Results